OpenShell

Purpose and PoC objective

- OpenShell is NVIDIA's policy-governed runtime for fleets of autonomous AI agents.

  Evidence: report.md

- This PoC tested whether its gateway, shared supervisor, and sandbox run on OpenShift under restricted-v2 with block storage.

  Evidence: report.md, plan.md

Testing strategy

- Five original scenarios covered gateway health, process execution, workspace persistence, and network allow/deny against a reachable control.

  Evidence: plan.md

- The denial was an expected policy outcome, measured with supervisor OCSF events; CNI, OIDC, HA, and load were excluded.

  Evidence: report.md, plan.md



Recorded setup

Two original checks

Health and sandbox execution



HEALTH + EXEC

200; Linux non-root

```
> health + exec        [+] 200; Linux non-root
persistence              [+] marker survived
network allow            [+] example.com 200
deny outcome           [-] blocked; 5/5 pass
```

Animation illustrates archived evidence; playback timing is illustrative.

ac05c639abd848998db29ce2352d9a0f
Report-based executive summary

How we tested

- The final original execution passed 5 of 5 scenarios; an earlier harness-only attempt is retained in the evidence.

  Evidence: test-results.json, attempt-index.json

- All three images were source-built with UBI; build times were 38m47s, 19m18s, and 13m27s with no build retries.

  Evidence: report.md, state.json



Validation results

5	0	0	0	0
pass	fail	error	skip	incomplete

- Single replica; no OIDC, HA, load, failover, filesystem, or wider-portability validation.
- Network denial is supervisor/proxy mediation; CNI enforcement was not independently tested.
- Google durable publication is blocked; no cloud links exist.

ac05c639abd848998db29ce2352d9a0f

Report-based executive summary



OpenShift and Kubernetes conclusions

- Gateway and sandbox workloads ran on the eu-gb-2 worker with 10Gi IBM block claims; restricted-v2 needed no privileged SCC.

  Evidence: report.md, state.json

- Tested compatibility is limited to this single-replica PoC; Kubernetes portability beyond it is inferred, not proven.

  Evidence: report.md

- Single replica; no OIDC, HA, load, failover, filesystem, or wider-portability validation.
- Network denial is supervisor/proxy mediation; CNI enforcement was not independently tested.
- Google durable publication is blocked; no cloud links exist.

Red Hat